Solution Architecture

Template: Solution Architecture Specification & Diagram	Maximum Marks: 5 Marks
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

PRESENTATION / CLIENT LAYER
• React 18 + Vite 6 Single-Page Application (Deployed on Vercel Global Edge CDN)
• HTML5 Canvas Synthetic Food Generator | Web Audio API Microphone Recorder
• Real-Time Interactive Telemetry & Status Monitoring Pill

↓ **(HTTPS / CORS / JSON Multipart Requests)** ↑

API GATEWAY & ROUTING LAYER
• FastAPI Asynchronous Server + Uvicorn ASGI Worker Pools (Hosted on Render Cloud)
• Pydantic v2 Strict Schema Validation & Sanitization | CORS Middleware (* Allowed)

↓ **(Asynchronous Microservice Routing)** ↑

CORE APPLICATION LOGIC SERVICES
• **Food Recognition Router:** Dimension normalization + Groq Multimodal vision analysis + Heuristic fallback
• **Freshness Detector Router:** OpenCV HSV discoloration masking + Laplacian gradient surface roughness
• **Hygiene Monitor Router:** Video frame sampler + Health audit reasoning (Grade A-F)
• **Kitchen Assistant Router:** Procedural recipe generator + Hands-free real-time cooking Q&A;
• **Meal Planner Router:** Mifflin-St Jeor metabolic target calculator + Multi-allergen exclusion engine

↓ **(External Model APIs & Local Storage)** ↑

DATA & EXTERNAL AI INFERENCE LAYER
• **Groq Cloud AI:** `qwen/qwen3.8-27b` Multimodal Vision & `whisper-large-v3-turbo`
• **Local Volatile Storage:** `temp_uploads/` scratch buffer with strict post-execution cleanup
• **Deterministic Heuristics Engine:** Context-aware fallback nutrition and recipe database

Component Description Table:

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Delivers responsive web interface, tabbed navigation, live testing workbench, and media capture.	React 18, Vite 6, CSS Variables, HTML5 Canvas API
API Gateway & Validation	Ingests client requests, validates input types, handles CORS, and routes to respective module controllers.	FastAPI 0.149, Pydantic v2, Python-Multipart
Computer Vision Engine	Standardizes images to 256x256, converts to HSV space, and computes discoloration index & Laplacian roughness.	OpenCV (opencv-python-headless), NumPy, PIL
Multimodal Reasoning	Performs plate decomposition, nutritional macro calculation, food safety inspection, and chef guidance.	Groq Cloud API (qwen/qwen3.8-27b), Whisper Turbo
Metabolic Planner	Calculates BMR and TDEE based on biometrics, distributes macros, and ensures strict allergen avoidance.	Python Mifflin-St Jeor Engine, Clinical Dietetics logic